

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. Examination (EE)

December 2011

EE 303 Microcontrollers & Their Applications

Date: 14.12.2011, Wednesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Explain the differences between Microcontrollers and Microprocessors. [04]
(b) Define the bits in PSW. [03]
- Q - 2 (a) What are the three buses in microcontroller based systems? State the purpose of each. [04]
(b) Draw and explain architecture block diagram of 8051. [06]
(c) Explain the internal 128 bytes of RAM structure of 8051? [04]

OR

- Q - 2 (a) Explain with example following addressing modes of 8051, [04]
Immediate, register, direct, register indirect, indexed.
(b) Draw the pin diagram of 8051 & also give the function of each pin. [06]
(c) What does "ROM" stand for? How is it used in computer systems? [04]
- Q - 3 (a) Explain the Timer/Counter with TCON and TMOD register. [04]
(b) Explain the stack and stack pointer of 8051. Explain its operation with related instructions. What is the maximum size of a stack in case of 8051? [04]
(c) A switch is connected to port pin P0.1. Write a program to check the status of the switch and perform the following: [06]
(a) If switch = 1, send a high-to-low pulse to activate a siren connected to pin P1.7.
(b) Continue monitoring the pin status.
Use carry flag to check the switch status.

OR

- Q - 3 (a) Explain different timer modes of 8051 Timer. [04]
(b) Explain the port (Port-0, Port-1, Port-2, and Port-3) structure (internal hardware) of 8051 with neat diagram for input and output configuration. [04]

[P.T.O.]

- (c) Write a program to perform the following, [06]
- Keep monitoring pin P0.1 until it becomes high
 - When P0.1 becomes high, read in the data from port 1
 - Send a low-to-high pulse on P0.2 to indicate that the data has been read.

SECTION – II

- Q - 4 (a) Mention the steps to create an executable assembly language program: How it is created, assembled and made ready to run? [04]
- (b) Describe CISC and RISC Architecture in short. [03]
- Q - 5 (a) What are the interrupts available in 8051? List out them and explain the IE register. [07]
- (b) Generate a square wave with an ON time of 3 ms and an OFF time of 10 ms on all pins of port 0. Assume an XTAL of 22 MHz. Use Timer 0 in Mode 1. [07]

OR

- Q - 5 (a) Two Switches are connected to pin P3.2 and P3.3. When a switch is pressed, the corresponding line goes low. Write a program to (a) light all LEDs connected to port 0, if the first switch is pressed. (b) Light all LEDs connected to port 2, if the second switch is pressed. [07]
- (b) Assuming that XTAL = 22 MHz, write a program to generate a square wave of frequency 1 kHz on pin P1.2. Use Timer 0 in Mode 2. [07]
- Q - 6 (a) Take data in through ports 0, 1 and 2, one after the other and transfer this data serially, continuously. Take crystal frequency of 11.0592 MHz and baud rate 4800. [08]
- (b) Draw the pin diagram of 8255 & also give the functionality of each pin. [06]

OR

- Q - 6 (a) What is the baud rate in 8051 and explain SBUF & SCON register. [06]
- (b) For Figure 1, (8255 interfacing with 8051) [08]
- Find the I/O port addresses assigned to ports A, B, C and control register.
 - Find the control byte for PA=in, PB=out, PC=out.
 - Write a program to get data from PA and send it to both ports B and C.

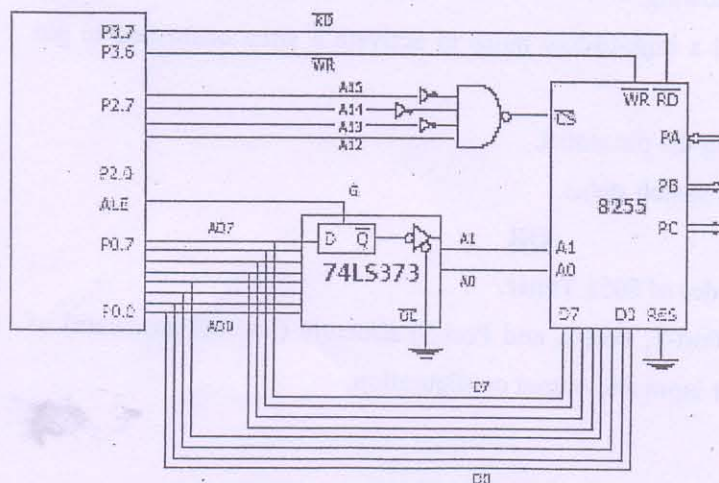


Figure 1